

CBSE Class 12 Maths — Half-Yearly Predicted Paper (Set 10 of 10)

SECTION A: Objective Type Questions (1 Mark Each x 20 = 20 Marks)

- Q1.** Find the total number of equivalence relations that can be defined on the set $A = \{1, 2, 3\}$. * **Chapter:** Relations and Functions * **Source:** [PYQ 2021] [Confidence: High]
- Q2.** Find the principal value of $\tan^{-1}(\sqrt{3}) - \sec^{-1}(-2)$. * **Chapter:** Inverse Trigonometric Functions * **Source:** [PYQ 2020] [Confidence: Very High]
- Q3.** If A is a 3×3 non-singular matrix such that $|A| = -3$, find the value of $|-A|$. * **Chapter:** Determinants * **Source:** [PYQ 2023] [Confidence: High]
- Q4.** Evaluate the definite integral: $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$. * **Chapter:** Integrals * **Source:** [PYQ 2018] [Confidence: Very High]
- Q5.** If a square matrix $A = [a_{ij}]$ is a skew-symmetric matrix of order 3, find the value of $a_{11} + a_{22} + a_{33}$. * **Chapter:** Matrices * **Source:** [PYQ 2019] [Confidence: High]
- Q6.** What is the derivative of x^x with respect to x ? * **Chapter:** Continuity and Differentiability * **Source:** [PYQ 2020] [Confidence: Very High]
- Q7.** Find the slope of the normal to the curve defined by $x = a \cos^3 \theta$ and $y = a \sin^3 \theta$ at $\theta = \frac{\pi}{4}$. * **Chapter:** Application of Derivatives * **Source:** [Practice] [Confidence: High]
- Q8.** Evaluate the indefinite integral: $\int \sec x (\sec x + \tan x) dx$. * **Chapter:** Integrals * **Source:** [PYQ 2022] [Confidence: Very High]
- Q9.** State the order of the differential equation: $\frac{d^2y}{dx^2} + \sin\left(\frac{dy}{dx}\right) = 0$. * **Chapter:** Differential Equations * **Source:** [PYQ 2021] [Confidence: High]
- Q10.** Write the integrating factor (I.F.) for the linear differential equation: $\frac{dy}{dx} - y = e^x$. * **Chapter:** Differential Equations * **Source:** [Practice] [Confidence: Very High]
- Q11.** Find the exact angle between the vector $\hat{i}$ and the vector $\hat{i} + \hat{j}$. * **Chapter:** Vector Algebra * **Source:** [Practice] [Confidence: Medium]
- Q12.** Calculate the perpendicular distance between the parallel planes $2x - y + 2z = 5$ and $2x - y + 2z = 8$. * **Chapter:** Three-Dimensional Geometry * **Source:** [PYQ 2019] [Confidence: Very High]
- Q13.** Find the scalar projection of the vector $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ on the vector $\vec{b} = \hat{i} - 2\hat{j} + 2\hat{k}$. * **Chapter:** Vector Algebra * **Source:** [PYQ 2020] [Confidence: Very High]
- Q14.** Write the Cartesian equation of the line passing through the point $(1, 2, 3)$ and parallel to the vector $\vec{v} = 2\hat{i} - \hat{j} + 3\hat{k}$. * **Chapter:** Three-Dimensional Geometry * **Source:** [PYQ 2018] [Confidence: High]
- Q15.** Find the positive value of x for which $\begin{vmatrix} 3 & x \\ x & 1 \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix}$. * **Chapter:** Determinants * **Source:** [PYQ 2023] [Confidence: High]
- Q16.** Find the principal value of $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$. * **Chapter:** Inverse Trigonometric Functions * **Source:** [Practice] [Confidence: High]
- Q17.** The radius of a circle is r . Find the rate of change of the area of the circle with respect to its radius when $r = 6$ cm. * **Chapter:** Application of Derivatives * **Source:** [PYQ 2019] [Confidence: Very High]
- Q18.** State the domain of the real-valued function $f(x) = \sqrt{x-1}$. * **Chapter:** Relations and Functions * **Source:** [Practice] [Confidence: Medium]
- Q19. (Assertion-Reason): Assertion (A):** $\int_{-a}^a \sin^3 x dx = 0$. **Reason (R):** For any odd continuous function $f(x)$, $\int_{-a}^a f(x) dx = 0$. Choose the correct option. * **Chapter:** Integrals * **Source:** [PYQ 2024] [Confidence: Very High]

Q20. (Assertion-Reason): Assertion (A): The vectors $2\hat{i} - \hat{j} + 2\hat{k}$ and $-4\hat{i} + 2\hat{j} - 4\hat{k}$ are collinear. **Reason (R):** Two vectors are collinear if and only if their cross product is the zero vector. Choose the correct option. * **Chapter:** Vector Algebra * **Source:** [Practice] [Confidence: High]

SECTION B: Very Short Answer Type Questions (2 Marks Each x 5 = 10 Marks)

Q21. Prove algebraically that $\sin^{-1}\left(\frac{3}{5}\right) - \sin^{-1}\left(\frac{8}{17}\right) = \cos^{-1}\left(\frac{84}{85}\right)$. * **Chapter:** Inverse Trigonometric Functions * **Source:** [PYQ 2019] [Confidence: Very High]

Q22. If $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$, verify that $A^2 - 4A + I = O$, where I is the identity matrix of order 2. * **Chapter:** Matrices * **Source:** [PYQ 2020] [Confidence: High]

Q23. If $y = a \cos(\log_e x) + b \sin(\log_e x)$, prove that $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$. * **Chapter:** Continuity and Differentiability * **Source:** [PYQ 2022] [Confidence: Very High]

Q24. Using differentials, find the approximate value of $\sqrt{0.037}$. * **Chapter:** Application of Derivatives * **Source:** [PYQ 2018] [Confidence: Medium]

Q25. Find the area of a parallelogram whose diagonals are uniquely determined by the vectors $\vec{d}_1 = \hat{i} + \hat{j} + \hat{k}$ and $\vec{d}_2 = \hat{i} - \hat{j} + \hat{k}$. * **Chapter:** Vector Algebra * **Source:** [PYQ 2021] [Confidence: High]

SECTION C: Short Answer Type Questions (3 Marks Each x 6 = 18 Marks)

Q26. Show that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 2x^3 - 5$ is a bijective function. * **Chapter:** Relations and Functions * **Source:** [PYQ 2019] [Confidence: Very High]

Q27. Using properties of determinants, prove that:

$$\begin{vmatrix} 1 + a^2 - b^2 & 2ab & -2b \\ 2ab & 1 - a^2 + b^2 & 2a \\ 2b & -2a & 1 - a^2 - b^2 \end{vmatrix} = (1 + a^2 + b^2)^3.$$

* **Chapter:** Determinants * **Source:** [PYQ 2020] [Confidence: Very High]

Q28. Differentiate the function $y = \sin^{-1}\left(\frac{2^{x+1}}{1+4^x}\right)$ with respect to x . * **Chapter:** Continuity and Differentiability * **Source:** [PYQ 2023] [Confidence: High]

Q29. Evaluate the following indefinite integral: $\int e^x \left(\frac{1 - \sin x}{1 - \cos x}\right) dx$. * **Chapter:** Integrals * **Source:** [PYQ 2019] [Confidence: Very High]

Q30. Solve the homogeneous differential equation: $(x^2 + xy)dy = (x^2 + y^2)dx$. * **Chapter:** Differential Equations * **Source:** [PYQ 2022] [Confidence: Very High]

Q31. Find the exact coordinates of the foot of the perpendicular drawn from the point $P(1, 3, 4)$ to the plane $2x - y + z + 3 = 0$. * **Chapter:** Three-Dimensional Geometry * **Source:** [PYQ 2020] [Confidence: High]

SECTION D: Long Answer Type Questions (5 Marks Each x 4 = 20 Marks)

Q32. Find the inverse of the matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix}$. Hence, use A^{-1} to solve the system of linear equations: $x - 2y = 10$, $2x - y - z = 8$, $-2y + z = 7$. * **Chapter:** Matrices and Determinants * **Source:** [PYQ 2023] [Confidence: Very High]

Q33. Show mathematically using calculus that the volume of the greatest right circular cylinder that can be inscribed in a given cone of height h and semi-vertical angle α is $\frac{4}{27}\pi h^3 \tan^2 \alpha$. * **Chapter:** Application of

Derivatives * **Source:** [PYQ 2018] [Confidence: Very High]

Q34. Evaluate the definite integral using standard properties of limits: $\int_0^\pi \log(1 + \cos x) dx$. * **Chapter:** Integrals * **Source:** [PYQ 2019] [Confidence: High]

Q35. Using the method of integration, sketch the region and find the exact area enclosed by the parabola $y^2 = 4ax$ and the line $y = mx$ (where $a > 0, m > 0$). * **Chapter:** Application of Integrals * **Source:** [Practice] [Confidence: Very High]

SECTION E: Case-Study Based Questions (4 Marks Each x 3 = 12 Marks)

Q36. Case Study 1: Matrices (Automotive Manufacturing) An automobile company manufactures three types of vehicles: cars, scooters, and bikes, operating out of two major assembly plants, Plant X and Plant Y. The monthly production array is modeled by a 2×3 matrix P . Plant X produces 100 cars, 200 scooters, and 50 bikes. Plant Y produces 120 cars, 150 scooters, and 100 bikes. The net profit per unit sold is ₹50,000 for a car, ₹10,000 for a scooter, and ₹5,000 for a bike, which is modeled by a column matrix Q . Assume all manufactured vehicles are sold. (i) Write the production matrix P and the profit matrix Q with their exact dimensions. (1 Mark) (ii) What matrix operation must be performed to calculate the total profit generated by each plant? Set up the expression. (1 Mark) (iii) Calculate the exact total profit generated by Plant X and Plant Y separately. (2 Marks) * **Chapter:** Matrices * **Source:** [Practice] [Confidence: High]

Q37. Case Study 2: Application of Derivatives (Architectural Arch) An architect is designing an aesthetic parabolic arch for a city bridge. The shape of the arch is defined on a coordinate plane by the mathematical equation $y = 9 - x^2$ (where x and y are in meters). The city council wants to hang a rectangular welcoming banner directly under this arch. The banner's base must lie perfectly flat on the x-axis, and its top two vertices must touch the inner curve of the parabola. Let the base vertices of the banner be $(-x, 0)$ and $(x, 0)$. (i) Express the total area $A(x)$ of the rectangular banner strictly as a function of x . (1 Mark) (ii) Find the first derivative $A'(x)$ and determine the critical point(s) for $x > 0$. (1 Mark) (iii) Use calculus to find the exact dimensions (width and height) of the banner that will maximize its surface area. (2 Marks) * **Chapter:** Application of Derivatives * **Source:** [Practice] [Confidence: Very High]

Q38. Case Study 3: Vector Algebra (Solar Panel Alignment) To maximize energy efficiency, a solar panel is mounted on a flat residential roof. Engineers model the three corners of the triangular solar panel as coordinates in a 3D space. The vertices are located at the points $A(2, 2, 0)$, $B(0, 4, 3)$, and $C(4, 1, 2)$ (measurements in meters). (i) Find the position vectors representing the structural edges $\vec{AB}$ and $\vec{AC}$. (1 Mark) (ii) Find a normal vector perpendicular to the surface of the solar panel by evaluating the cross product $\vec{AB} \times \vec{AC}$. (1 Mark) (iii) Calculate the exact total surface area of this triangular solar panel. (2 Marks) * **Chapter:** Vector Algebra * **Source:** [Practice] [Confidence: High]

Answer Key

Q1. 5. (The equivalence relations correspond to the partitions of the set). **Q2.**

$\tan^{-1}(\sqrt{3}) - \sec^{-1}(-2) = \frac{\pi}{3} - \frac{2\pi}{3} = -\frac{\pi}{3}$. **Q3.** $|-A| = (-1)^n |A|$. For a 3×3 matrix,

$|-A| = (-1)^3 (-3) = 3$. **Q4.** $\int_0^1 \frac{dx}{\sqrt{1-x^2}} = [\sin^{-1} x]_0^1 = \sin^{-1}(1) - \sin^{-1}(0) = \frac{\pi}{2}$. **Q5.** In a skew-symmetric matrix, the principal diagonal elements are zero. Thus, $0 + 0 + 0 = 0$. **Q6.** Let

$y = x^x \implies \log y = x \log x \implies \frac{1}{y} \frac{dy}{dx} = x \left(\frac{1}{x} \right) + \log x(1) \implies \frac{dy}{dx} = x^x (1 + \log x)$. **Q7.**

$\frac{dx}{d\theta} = -3a \cos^2 \theta \sin \theta$, $\frac{dy}{d\theta} = 3a \sin^2 \theta \cos \theta$. $\frac{dy}{dx} = \frac{3a \sin^2 \theta \cos \theta}{-3a \cos^2 \theta \sin \theta} = -\tan \theta$. At $\theta = \pi/4$, slope of tangent is -1 . Slope of normal is 1 . **Q8.** $\int (\sec^2 x + \sec x \tan x) dx = \tan x + \sec x + C$. **Q9.** Order is 2 (highest

derivative is $d^2 y/dx^2$). **Q10.** I.F. $= e^{\int -1 dx} = e^{-x}$. **Q11.** $\cos \theta = \frac{i \cdot (i+j)}{|i||i+j|} = \frac{1}{(1)(\sqrt{2})} = \frac{1}{\sqrt{2}} \implies \theta = \frac{\pi}{4}$. **Q12.**

Distance $= \frac{|d_1 - d_2|}{\sqrt{A^2 + B^2 + C^2}} = \frac{|8 - 5|}{\sqrt{2^2 + (-1)^2 + 2^2}} = \frac{3}{\sqrt{9}} = \frac{3}{3} = 1$ unit. **Q13.** Projection

$$= \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{(2)(1) + (-1)(-2) + (1)(2)}{\sqrt{1^2 + (-2)^2 + 2^2}} = \frac{2+2+2}{\sqrt{9}} = \frac{6}{3} = 2. \text{ Q14. } \frac{x-1}{2} = \frac{y-2}{-1} = \frac{z-3}{3}. \text{ Q15.}$$

$$3 - x^2 = 3 - 8 \implies x^2 = 8 \implies x = 2\sqrt{2} \text{ (taking positive value). Q16. } \sin^{-1}(1/\sqrt{2}) = \frac{\pi}{4}. \text{ Q17.}$$

$A = \pi r^2 \implies \frac{dA}{dr} = 2\pi r$. At $r = 6$, rate is $12\pi \text{ cm}^2/\text{cm}$. Q18. We require $x - 1 \geq 0 \implies x \geq 1$. Domain is $[1, \infty)$. Q19. Option A. (Both A and R are true, and R is the correct explanation.

$f(-x) = \sin^3(-x) = -\sin^3 x = -f(x)$. Q20. Option A. (Both A and R are true, and R is the correct explanation. $\vec{a} \times \vec{b} = \vec{0}$ is the condition for collinearity).

Q21. Let $\alpha = \sin^{-1}(3/5) \implies \sin \alpha = 3/5, \cos \alpha = 4/5$. Let

$\beta = \sin^{-1}(8/17) \implies \sin \beta = 8/17, \cos \beta = 15/17$.

$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta = (4/5)(15/17) + (3/5)(8/17) = 60/85 + 24/85 = 84/85$. Hence

proved. Q22. $A^2 = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 12 \\ 4 & 7 \end{bmatrix}$.

$$A^2 - 4A + I = \begin{bmatrix} 7 & 12 \\ 4 & 7 \end{bmatrix} - \begin{bmatrix} 8 & 12 \\ 4 & 8 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O. \text{ Q23.}$$

$y' = -a \sin(\log x) \cdot \frac{1}{x} + b \cos(\log x) \cdot \frac{1}{x} \implies xy' = -a \sin(\log x) + b \cos(\log x)$. Differentiate again:

$xy'' + y' = -a \cos(\log x) \cdot \frac{1}{x} - b \sin(\log x) \cdot \frac{1}{x} \implies x^2 y'' + xy' = -(a \cos(\log x) + b \sin(\log x)) = -y$.

Thus $x^2 y'' + xy' + y = 0$. Q24. Let $f(x) = \sqrt{x}$. Take $x = 0.04, \Delta x = -0.003$.

$$dy = \frac{1}{2\sqrt{x}} \Delta x = \frac{1}{2(0.2)} (-0.003) = \frac{-0.003}{0.4} = -0.0075. \text{ Approx value} = f(x) + dy = 0.2 - 0.0075 = 0.1925$$

$$\text{Q25. Area} = \frac{1}{2} |\vec{d}_1 \times \vec{d}_2|. \vec{d}_1 \times \vec{d}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{vmatrix} = \hat{i}(1 - (-1)) - \hat{j}(1 - 1) + \hat{k}(-1 - 1) = 2\hat{i} - 2\hat{k}.$$

Magnitude $= \sqrt{4 + 0 + 4} = \sqrt{8} = 2\sqrt{2}$. Area $= \frac{1}{2} (2\sqrt{2}) = \sqrt{2}$ sq units.

$$\text{Q26. One-one: } f(x_1) = f(x_2) \implies 2x_1^3 - 5 = 2x_2^3 - 5 \implies 2x_1^3 = 2x_2^3 \implies x_1^3 = x_2^3 \implies x_1 = x_2.$$

Onto: Let $y = 2x^3 - 5 \implies x^3 = \frac{y+5}{2} \implies x = \left(\frac{y+5}{2}\right)^{1/3}$. Since for every $y \in \mathbb{R}, x \in \mathbb{R}$ exists, it is onto.

Thus, bijective. Q27. Apply $C_1 \rightarrow C_1 - bC_3$ and $C_2 \rightarrow C_2 + aC_3$. C_1 becomes

$(1 + a^2 + b^2), 0, b(1 + a^2 + b^2)$. C_2 becomes $0, (1 + a^2 + b^2), -a(1 + a^2 + b^2)$. Take $(1 + a^2 + b^2)$

common from C_1 and C_2 . Expanding the remaining determinant yields the third $(1 + a^2 + b^2)$ factor. Q28. Let

$$2^x = \tan \theta. y = \sin^{-1} \left(\frac{2 \cdot 2^x}{1 + (2^x)^2} \right) = \sin^{-1} \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right) = \sin^{-1}(\sin 2\theta) = 2\theta = 2 \tan^{-1}(2^x).$$

$$\frac{dy}{dx} = \frac{2}{1 + (2^x)^2} \cdot \frac{d}{dx}(2^x) = \frac{2}{1 + 4^x} \cdot 2^x \log_e 2 = \frac{2^{x+1} \log_e 2}{1 + 4^x}. \text{ Q29.}$$

$$\int e^x \left(\frac{1 - 2 \sin(x/2) \cos(x/2)}{2 \sin^2(x/2)} \right) dx = \int e^x \left(\frac{1}{2} \csc^2(x/2) - \cot(x/2) \right) dx. \text{ Let}$$

$$f(x) = -\cot(x/2) \implies f'(x) = \frac{1}{2} \csc^2(x/2). \text{ The integral is}$$

$$\int e^x [f(x) + f'(x)] dx = e^x f(x) + C = -e^x \cot(x/2) + C. \text{ Q30. } \frac{dy}{dx} = \frac{x^2 + y^2}{x^2 + xy}. \text{ Put}$$

$$y = vx \implies \frac{dy}{dx} = v + x \frac{dv}{dx}. v + x \frac{dv}{dx} = \frac{1+v^2}{1+v} \implies x \frac{dv}{dx} = \frac{1+v^2-v-v^2}{1+v} = \frac{1-v}{1+v}. \text{ Separating variables:}$$

$$\int \frac{1+v}{1-v} dv = \int \frac{dx}{x} \implies \int (-1 + \frac{2}{1-v}) dv = \log x + C \implies -v - 2 \log |1-v| = \log x + C. \text{ Resubstitute}$$

$v = y/x$. Q31. Line through $(-1, 4, 3)$ perpendicular to plane has DRs $2, -1, 1$. Equation:

$$\frac{x+1}{2} = \frac{y-4}{-1} = \frac{z-3}{1} = k. \text{ Coordinates of foot } Q(2k-1, -k+4, k+3). Q \text{ lies on plane:}$$

$$2(2k-1) - (-k+4) + (k+3) + 3 = 0 \implies 4k - 2 + k - 4 + k + 3 + 3 = 0 \implies 6k = 0 \implies k = 0$$

. The foot is $(-1, 4, 3)$.

Q32. Let $A^T X = B \implies X = (A^T)^{-1} B = (A^{-1})^T B$. Alternatively, write the system as

$$\begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix}^T \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 8 \\ 7 \end{bmatrix}. \text{ Solving step by step: } |A| = 1(-1-2) - 2(-2-0) = -3 + 4 = 1 \neq 0.$$

Find $\text{adj} A$ and A^{-1} . Multiplying gives $x = 2, y = -1, z = 3$. Q33. Let cylinder radius be R and height be y .

From similar triangles, $\frac{h-y}{h} = \frac{R}{r} \implies y = h(1 - \frac{R}{r})$. Volume $V = \pi R^2 y = \pi h(R^2 - \frac{R^3}{r})$.

$$V'(R) = \pi h(2R - \frac{3R^2}{r}) = 0 \implies R = \frac{2r}{3}. \text{ Since } r = h \tan \alpha, R = \frac{2h \tan \alpha}{3}. y = h/3. \text{ Max volume}$$

$$= \pi \left(\frac{4}{9} h^2 \tan^2 \alpha \right) \left(\frac{h}{3} \right) = \frac{4}{27} \pi h^3 \tan^2 \alpha. \text{ Q34. Let } I = \int_0^\pi \log(1 + \cos x) dx. I = \int_0^\pi \log(1 - \cos x) dx. \text{ Adding}$$

$$\text{both: } 2I = \int_0^\pi \log(1 - \cos^2 x) dx = \int_0^\pi \log(\sin^2 x) dx = 2 \int_0^\pi \log(\sin x) dx.$$

$I = 2 \int_0^{\pi/2} \log(\sin x) dx = 2(-\frac{\pi}{2} \log 2) = -\pi \log 2$. **Q35.** Intersections of $y^2 = 4ax$ and $y = mx \implies (mx)^2 = 4ax \implies m^2x^2 = 4ax \implies x = 0, x = 4a/m^2$. Area $= \int_0^{4a/m^2} (\sqrt{4ax} - mx) dx = [2\sqrt{a} \frac{x^{3/2}}{3/2} - \frac{mx^2}{2}]_0^{4a/m^2} = \frac{4\sqrt{a}}{3} (\frac{4a}{m^2})^{3/2} - \frac{m}{2} (\frac{4a}{m^2})^2 = \frac{32a^2}{3m^3} - \frac{8a^2}{m^3} = \frac{8a^2}{3m^3}$ sq units.

Q36. (i) $P = \begin{bmatrix} 100 & 200 & 50 \\ 120 & 150 & 100 \end{bmatrix}$ (Dimensions: 2×3). $Q = \begin{bmatrix} 50000 \\ 10000 \\ 5000 \end{bmatrix}$ (Dimensions: 3×1). (ii) Total Profit Matrix

$= P \times Q$. (iii)

$$P \times Q = \begin{bmatrix} 100(50000) + 200(10000) + 50(5000) \\ 120(50000) + 150(10000) + 100(5000) \end{bmatrix} = \begin{bmatrix} 5000000 + 2000000 + 250000 \\ 6000000 + 1500000 + 500000 \end{bmatrix} = \begin{bmatrix} 7250000 \\ 8000000 \end{bmatrix}.$$

Plant X Profit = ₹72,50,000; Plant Y Profit = ₹80,00,000. **Q37.** (i) Width is $2x$, height is $y = 9 - x^2$. Area

$A(x) = 2x(9 - x^2) = 18x - 2x^3$. (ii) $A'(x) = 18 - 6x^2$. Set

$A'(x) = 0 \implies 6x^2 = 18 \implies x^2 = 3 \implies x = \sqrt{3}$ (since $x > 0$). (iii) $A''(x) = -12x$. $A''(\sqrt{3}) < 0$ (Max). Width = $2\sqrt{3}$ m. Height = $9 - (\sqrt{3})^2 = 6$ m. Dimensions: $2\sqrt{3}$ m by 6 m. **Q38.** (i)

$$\vec{AB} = (0 - 2)\hat{i} + (4 - 2)\hat{j} + (3 - 0)\hat{k} = -2\hat{i} + 2\hat{j} + 3\hat{k}.$$

$$\vec{AC} = (4 - 2)\hat{i} + (1 - 2)\hat{j} + (2 - 0)\hat{k} = 2\hat{i} - \hat{j} + 2\hat{k}. \text{ (ii) Normal}$$

$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 2 & 3 \\ 2 & -1 & 2 \end{vmatrix} = \hat{i}(4 - (-3)) - \hat{j}(-4 - 6) + \hat{k}(2 - 4) = 7\hat{i} + 10\hat{j} - 2\hat{k}. \text{ (iii) Area}$$

$$= \frac{1}{2} |\vec{AB} \times \vec{AC}| = \frac{1}{2} \sqrt{49 + 100 + 4} = \frac{1}{2} \sqrt{153} \text{ sq meters.}$$

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